Two microstructures for *Competition Creates Competition*: results, trade-offs and proof method

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This note compares two trading stages for the model of *Competition Creates Competition*. In the first, the Laplace set-up, an investor chooses a continuous order and Laplace noise hides it. In the second, the order-slot set-up, the investor shares one unit order slot with a liquidity trader. The auction, the preparation costs and the challenger's decision are the same in both.

Sections 1 to 3 state the set-ups, the results side by side, and the trade-offs as results with their conditions. Section 4 sets out the proof method and works one proof in each set-up. Section 5 records where each manuscript stands against that method, and Section 6 lists decision points.

The Laplace manuscript is the version of 22 October 2026 (main text, Appendix A and the Online Appendix). The slot manuscript is complete: Sections 1 to 9 and Appendix A.1 to A.9. Slot citations use its numbering: Propositions 1 to 4 in the main text, Propositions A.1 to A.14 in the appendix. Every slot number comes from the project registry. Every Laplace number comes from an independent outward recomputation of its benchmark or from its printed statements. Each result carries one status label: analytical, computer-assisted, numerical diagnostic or open.

Symbols follow each manuscript. Where they collide, the note writes the slot's research bound as r_U^R and keeps r_U for the Laplace full-order bound.

1 The two set-ups

Shared primitives. Both set-ups keep Sections 2.1, 2.2 and 3.1 of the Laplace manuscript. A challenger of class $j \in \{H, L\}$, with prior $\pi = \Pr(j = H)$, has a value $\theta \sim G_j$. The class G_H dominates G_L, and both supports lie above the reserve p. The incumbent's value is $R \sim F$, uniform on $[0, r]$ in the benchmark. The challenger's preparation cost is 0 with probability ρ and c_H otherwise. Prepared bidders bid their values in a second-price auction with reserve p.

The auction objects are t_0, t_j, g_j and $\Delta_T = t_H - t_L$. At a belief μ the challenger's gross profit is $B(\mu) = g_L + \mu(g_H - g_L)$, and the threshold belief is $\tau = (c_H - g_L)/(g_H - g_L)$. A stronger incumbent lowers g_H and g_L and raises Δ_T (Proposition 1 of either manuscript; analytical). The benchmark auction is $(h, \ell, p, \rho, c_H, k) = (10, 1, 1/2, 1/4, 6, 1/50)$, with prior $1/2$ and strengths $r_0 = 6/5$, $r_1 = 3$, $r_2 = 18/5$. Entry E is the probability that the challenger prepares, and high-class ownership O_H the probability that a class-H challenger acquires the target.

1.1 The Laplace set-up

Primitives and timing. The investor observes the class and submits an order $q \in [-1, 1]$ at the linear cost $k|q|$. Noise demand Z has the Laplace density $f(z) = e^{-|z|/b}/(2b)$ with scale $b > 1$. Market makers see the flow $X = q + Z$ and set $P(X) = \mathbb{E}[V_T \mid X]$, where V_T is the terminal payoff of a target share (Section 2.3). The timing is the sale rule, the investor's trade, the pricing of the flow, the challenger's decision and the auction.

What the challenger learns from the price. For any two orders, the likelihood ratio of a flow lies in $[e^{-2/b}, e^{2/b}]$. Under every mixed order profile the posterior after every flow therefore lies in $[m(\pi), M(\pi)]$, with $m(\pi) = \pi/[\pi + (1-\pi)e^{2/b}]$ and $M(\pi) = \pi/[\pi + (1-\pi)e^{-2/b}]$ (Proposition A.1; analytical). Every flow has a positive density in both classes, so every price is on path. The zero-cost challenger prepares after every price, so entry is at least ρ. Competitive pricing gives $P = t_0 + e(P)[t_L - t_0 + \Delta_T \mu_X]$, and the challenger inverts this identity to recover μ_X whenever $\Delta_T > 0$ (Proposition A.2; analytical).

The investor's residual advantage at a flow is $A_H = e\Delta_T(1-\mu_X)$ in class H and $A_L = e\Delta_T\mu_X$ in class L. Both lie in $[\rho\kappa(\pi)\Delta_T, \Delta_T]$, with $\kappa(\pi) = \min\{m(\pi), 1-M(\pi)\}$ (equation (11)). Through the score bound $|f'| \le f/b$, a correctly signed order of size s has marginal profit at least $(1 - s/b)\bar{A} - k$ (Appendix A.2).

Calibration. The benchmark uses $b = 2$, so $M(1/2) = e/(1+e) = 0.731059$ and $m(1/2) = 0.268941$. Entry is 0.250000 at r_0, 0.522757 at r_1 and 0.250000 at r_2. High-class ownership is 0.125000, 0.324192 and 0.125000 (Proposition 2 and Table 2; analytical).

1.2 The order-slot set-up

Primitives and timing. One public order opportunity comes before the challenger's decision. With probability $\lambda \in (0, 1)$ the informed investor fills it. It sees the class and chooses $q \in \{-1, 0, 1\}$, and a nonzero order costs k. With probability $1 - \lambda$ a liquidity trader fills the slot. It sells, abstains or buys one unit with probabilities β, $1 - 2\beta$, β, where $0 < \beta \le 1/3$, and its motive is exogenous. No one observes who filled the slot.

Market makers see the order $X \in \{-1, 0, 1\}$ and set $P = \mathbb{E}[V_T \mid X]$ (Section 2.3). The timing is that of the Laplace set-up, with the slot draw at the trading date; Section 5 adds a research stage before this date. The device is the alternative-arrival structure of Dow and Gorton (1997); the unit menu, the presence probability and the unit cost follow Edmans et al. (2015).

What the challenger learns from the price. Two numbers summarize the slot. The first is $b = (1 - \lambda)\beta$, the probability of a liquidity order in one direction. The second is $u = \lambda/b$, the informed presence mass relative to the liquidity mass in one direction. The largest likelihood ratio that one order can carry is $1 + u$. Under every investor strategy the posterior after a nonzero order lies in $[m(\pi), M(\pi)]$, with $M(\pi) = \pi(1+u)/[\pi(1+u) + 1 - \pi]$ and $m(\pi) = \pi/[\pi + (1-\pi)(1+u)]$ (Lemma A.1; analytical). The bound at the zero order needs $\beta \le 1/3$, and no other result needs it.

Every order has positive probability in both classes, so every price is on path. Equal prices give equal entry and hence equal posteriors, so the price reveals the posterior (Lemma A.2). Each

Table 1. The two set-ups.

	Laplace	Order slot
Order menu	$q \in [-1, 1]$, cost $k\|q\|$	$q \in \{-1, 0, 1\}$, cost k per nonzero order
What hides the order	additive Laplace noise Z, scale $b > 1$	a liquidity trader fills the slot with probability $1 - \lambda$, orders $(\beta, 1 - 2\beta, \beta)$
What market makers see	the flow $X = q + Z \in \mathbb{R}$	the order $X \in \{-1, 0, 1\}$
Largest likelihood ratio	$e^{2/b}$	$1 + u$, $u = \lambda/[(1 - \lambda)\beta]$
Posterior bounds	$[m(\pi), M(\pi)]$ at every flow	$[m(\pi), M(\pi)]$ at every order; the zero order needs $\beta \leq 1/3$
Price sufficiency	inversion through the entry floor, $\Delta_T > 0$	equal prices give equal posteriors, $\Delta_T > 0$
Investor's problem	a derivative bound on $[0, 1]$: $U_j'(s) \geq (1 - 1/b)\rho\kappa\Delta_T - k$	one comparison per class: K_j against k
Trade bound in Proposition 2	$k < (1 - 1/b)\rho\kappa\Delta_T(F_1)$	$k < \rho\kappa\Delta_T(F_1)$
Added assumptions	a continuous order; a Laplace law with $b > 1$ (Proposition A.5 extends to every bounded-score density)	a unit menu; a presence probability; an exogenous liquidity motive; with research, a quadratic cost and a cap
Tie rules	challenger prepares at indifference; no investor rule needed in Propositions 2 and 3	investor IM (any mixture over maximizers), challenger C1; I0 and CM defined in Appendix A.4 and stated where they change a result

class compares one gross return with k. A purchase earns $K_H - k$ with $K_H = e(1)\Delta_T[1 - \mu(1)]$ in class H, and a sale earns $K_L - k$ with $K_L = e(-1)\Delta_T\mu(-1)$ in class L. Under every candidate, $\rho\kappa(\pi)\Delta_T \leq K_j \leq \Delta_T$ (Lemma A.3; analytical).

Calibration B. The slot has two primitives: $1 + u = e$ and $\beta = 0.294655$, an exact decimal. Then $\lambda = u\beta/(1 + u\beta) = 0.336122$ and $b = 0.195615$. With $1 + u = e^{2/b}$ at the Laplace scale $b = 2$, the posterior bounds of the two set-ups coincide: $M(1/2) = 0.731059$ and $m(1/2) = 0.268941$. Entry is 0.250000, 0.522757 and 0.250000 at the three strengths. High-class ownership is 0.125000, 0.324401 and 0.125000 (Proposition 2 and Table 3; computer-assisted).

In Table 1, the posterior bounds, the entry floor, the price sufficiency and the residual bounds have the same form in both set-ups. The investor's problem differs. It is an integral derivative bound over a continuum of orders in the Laplace set-up, and one comparison per class in the slot. The slot's trade bound has no factor $1 - 1/b$, because a unit order changes no density. Neither set-up needs an off-path belief.

2 Main results side by side

Table 2 lists each result with its statement and status in the two manuscripts. References are to the propositions, corollaries, tables and sections of each manuscript.

Table 2. Main results in the two manuscripts.

Result	Laplace: statement and status	Slot: statement and status
Payoffs and target value	Proposition 1: a strengthening lowers g_H, g_L and raises Δ_T. Analytical.	Proposition 1: the same proposition and proof. Analytical.
Posterior bounds under every strategy	Proposition A.1: $m(\pi) \le \mu \le M(\pi)$ at every flow, from the score bound. Analytical.	Lemma A.1: the same bounds at every nonzero order; at the zero order when $\beta \le 1/3$. Analytical.
Price sufficiency	Proposition A.2: with $\Delta_T > 0$ the price reveals μ_X; residual bounds (11). Analytical.	Lemma A.2: equal prices give equal posteriors; prices rise strictly with the posterior. Lemma A.3: one comparison per class. Analytical.
Proposition 2 (i)–(iii)	Unique no trade at F_0; unique full orders at F_1 (entry above ρ) and at F_2 (entry ρ), against arbitrary mixed orders on $[-1, 1]$. Conditions (A1), (A2) with the factor $1 - 1/b$, and $B_{F_2}(M) < c_H$. Analytical.	Proposition 2: the same three outcomes, unique on the unit menu, the same under all four tie-rule pairs. Conditions (H1)–(H3) without the factor $1 - 1/b$; ownership rises iff $[\rho + (1-\rho)(b+\lambda)]W_H(F_1) > \rho W_H(F_0)$. Analytical.
Admissible region	Section 4.2: open in (r_0, r_1, c_H, k), nonempty (Appendix A.4); prior windows $(0.467850, 0.538096)$ for (i)–(ii) and $(0.467850, 0.500424)$ for all three parts; cost window $(5.997311, 6.217155)$. Analytical; not tabulated.	Proposition A.1 and Table 2: the exact band $82/59 < u < 2686/1559$, that is $\tau(r_1) < M(1/2) < \tau(r_2)$; prior windows $(0.467850, 0.633117)$ and $(0.467850, 0.500424)$; tabulated with margins. Analytical.
Fixed-information control	Proposition A.3: at a fixed experiment entry falls weakly. Benchmark: full orders held fixed, entry falls from 0.562178 to 0.522757 (closed forms; Table 2 labels the rows numerical diagnostic).	Proposition A.3: entry does not rise under rule C1, for every experiment whose signal is independent of the incumbent's value given the class, $\mathcal{L}_i(\theta, R \mid j, S) = G_j \otimes F_i$, and every independent cost law. With full orders held fixed it is unchanged at 0.522757 on $(1, r_C]$ and ρ above r_C (Table 3). Analytical; values computer-assisted.
Coexistence at the benchmark	Proposition 3: equilibria $(1, -v)$ at 1.55, 1.60, 1.65 with entry 0.545053, 0.548756, 0.551361, strictly ordered, each beside no trade. Computer-assisted at three nodes; the branch between them is a numerical diagnostic, and the full correspondence is open.	Proposition 3: the complete equilibrium set on $[r_0, r_2]$ at prior $1/2$ for every slot in the band. Coexistence exactly on $[r_A, r_N] = [1.393958, 1.747878]$; entry rises strictly on $[r_A, r_F]$, $r_F = 1.467127$, and is flat after. At 1.55 to 1.65 entry is 0.522757 beside no trade. Computer-assisted (two Sturm counts).
Coexistence at other priors	Not stated.	Corollary A.2: the coexistence set at every prior. Under C1 it is empty below $\pi_A^{\mathrm{lo}} \approx 0.396175$ and from $\pi_{\max}$ on, with $\pi_{\max} \approx 0.650059$; it starts at r_A up to $\pi_A^{\mathrm{hi}} \approx 0.640739$. Under CM one point remains at $\pi_{\max}$, with margin 0.000343 (Appendix A.4). Computer-assisted.
Research	Not modeled; Section 8 names endogenous research as a next step.	Proposition 4, Proposition A.2: zero research unique at r_0; a research branch on $[1.501023, 1.677633)$ with entry $0.507630 < 0.513965 < 0.519763$ at 1.55, 1.60, 1.65, beside zero research; research at the cap unique at r_1 and r_2. Slope analytical; lists and values computer-assisted.

Result	Laplace: statement and status	Slot: statement and status
Preparation cost laws	Proposition A.6: atomless costs keep (i)–(ii); entry 0.522715 at half-width 0.1. Analytical.	Proposition A.4: any independent cost law with the stated supports, atoms allowed; entry 0.522757; the return at r_2 needs a half-width below 0.002688902 (entry 0.382711 at 0.1). Analytical; values computer-assisted.
Value classes and priors	Proposition A.5 and Appendix A.4: a nonempty open set for every ranked pair; Table 4 cases A to C hold and D fails (A1) by -0.018122. Analytical.	Proposition A.5 and Table A.4: the same for every ranked pair and prior; every Table 4 case with its own costs; D is a control. Analytical; values computer-assisted.
Many classes	Online Appendix A.13: the price reveals the full posterior; two groups under a separation $\gamma > 0$. Analytical.	Proposition A.6: two groups under separation; a strict three-class equilibrium where two orders share one price (knife-edge). Analytical; example computer-assisted.
Complementary signals	Proposition A.7 at vector (A.50): entry $0.850000 \to 0.879438$. Analytical.	Proposition A.7: the same vector, $0.850000 \to 0.879797$; at the benchmark, accuracies $(0.94, 0.505)$ give $0.250000 \to 0.386933$, with no rise when the challenger's signal is removed. Analytical; values computer-assisted.
Access to prices	Proposition A.9: proceeds +0.257809, net surplus +0.080218 at r_1. Analytical. Break-even trade scale 1.831923 (numerical diagnostic).	Proposition A.8: +0.257948 and +0.081960; the same with research at the cap. Break-even scale 6.848394. Analytical; values computer-assisted.
Payment rule	Proposition A.8: the spread response has the sign of $1 - 2\omega$ (η in the manuscript). Analytical.	Proposition A.9: the same, weakly; strictly when $\mathbb{E}[(R - \ell)_+]$ changes strictly and $\omega \neq 1/2$. Analytical.
Reserves	Table 5: at the reserve 1.01 preparation falls from 0.543573 to 0.513373; unique continuations at each node. Analytical.	Proposition A.10 and Table 5: at the reserve 101/100 entry is 0.522757 at both strengths; the reserve 31/20 restores strict deterrence and excludes the weak incumbent. Analytical; values computer-assisted.
Empirical implications	Section 7: three predictions for visible sale processes; no observable is derived.	Section 8, Propositions A.11 to A.14: the conditional execution spread $S(r_1) = 0.786299$ with entry share 90.2%, zero above r_C; identification of $(\vartheta, \lambda, \beta)$ from sell impact and volume; orders predict the winner above r_C; announcement returns. Analytical; values computer-assisted.

Three facts organize the table. First, the reversal and its welfare consequence are the same results in both set-ups. Calibration B gives the slot the Laplace top posterior, so every margin that concerns the auction alone is the same number. The cost window $(5.997311, 6.217155)$ and the three-part prior window $(0.467850, 0.500424)$ are identical. Second, the slot replaces three certified points with a complete equilibrium set, and it adds research, coexistence at every prior and derived observables. Third, two comparative statements lose their strict form in the slot: the fixed-information control and reserve deterrence at the exclusion reserve.

Uniqueness has a different scope in the two set-ups. In the Laplace set-up it covers arbitrary mixed orders on a continuum and every unilateral deviation in $[-1, 1]$. It is proved at the three benchmark strengths and on the two outer regions, $r < \mathfrak{r}(k) = 1.220998$ and $r > r_U = 2.837417$.

In the slot it covers arbitrary mixtures on three orders. The whole set is known at every strength on the path (Proposition 3). Full orders are unique on $r_S < r < h$, where $r_S = 2.124148$ is the exact root of $\rho\Delta_T m = k$ (Section 8). The second is a weaker uniqueness claim per economy and a stronger completeness claim along the path.

3 Trade-offs

Each trade-off below is a result with its condition. Section 3.1 concerns the dynamics each set-up generates. Section 3.2 concerns the conclusions each supports.

3.1 Dynamics

The partial sale. *Laplace.* At 1.55, 1.60 and 1.65 the certified equilibria have pure orders $(1, -v)$, with $v = 0.460316$, 0.707475 and 0.903332 (Proposition 3; computer-assisted). The interior sale exists because the low class's marginal profit falls strictly with its order (Online Appendix B.2). A root of its first-order condition is therefore a global optimum. The branch between the nodes is a numerical diagnostic.

Slot. On the unit menu a sale is all or nothing, and the only continuous margin is a mixture probability. The low class abstains up to $r_N = 1.747878$, mixes on (r_N, r_S) and sells with probability one from $r_S = 2.124148$ (Proposition 3; computer-assisted). The mixture needs the investor rule IM. Under I0 the equilibrium set is empty on $(r_N, r_S]$ (Appendix A.4).

The partial sale is therefore a property of the continuous order menu. Its condition is that the investor can choose a size, with a density that moves smoothly with it.

Strictly rising entry at intermediate strengths. *Laplace.* Entry rises strictly across the three nodes, $0.545053 < 0.548756 < 0.551361$, because the entry enclosures are disjoint (Proposition 3; computer-assisted). The sale magnitude carries the rise: a larger v makes a sale more informative and moves the threshold flow. The conditions are the continuous menu, a fixed presence of the investor and an interval certificate at each node.

Slot. With a fixed presence probability, entry rises strictly on $[r_A, r_F] = [1.393958, 1.467127]$, where class H mixes (Proposition 3(b); computer-assisted). Entry is 0.454185 at r_A, 0.506112 at 1.45 and 0.522757 from r_F. The purchase rate $s_+ = b[\Delta_T(r)/k - 2]$ adjusts so that a purchase that recruits earns exactly k. The rise needs rule IM, and entry is flat on $[r_F, r_C]$; at 1.55 to 1.65 it equals 0.522757.

Slot. With research, entry rises strictly on $[1.501023, 1.677633)$, which contains the three strengths: $0.507630 < 0.513965 < 0.519763$ (Proposition 4; slope analytical, values computer-assisted). The research intensity carries the rise. It solves $2a\lambda = \Delta_T(r)m_\lambda - k$, whose two sides meet once, and class H buys with probability one after a success. The conditions are a quadratic research cost and a cap in the strip $(0.290539, 0.336721)$; at the benchmark cap the cost lies in $(0.021469, 0.036763)$ (Proposition A.2). Zero research is an equilibrium at every strength on the branch.

Each set-up therefore has a strict rise at intermediate strengths. A different object carries it in each case: a sale size, a mixture probability or a research intensity. The slot's fixed-presence rise

sits at lower strengths than the Laplace nodes. Its research rise sits at the Laplace nodes with lower entry levels.

Coexistence. *Laplace.* No trade is an equilibrium exactly on $r \le r_N = 1.747878$ and is unique below $\mathfrak{r}(k) = 1.220998$. Full orders are unique above $r_U = 2.837417$ (Proposition A.4; analytical). Between the two bounds the manuscript certifies three informative equilibria, each beside no trade, and proves no uniqueness. The equilibrium set there is open.

Slot. No trade and an informative equilibrium coexist exactly on $[r_A, r_N]$, and the set at each strength is listed (Proposition 3; computer-assisted). Under rule CM the threshold pair is also an equilibrium on $(r_A, r_J]$, $r_J = 1.966097$, with entry that falls strictly from 0.454185 to ρ (Appendix A.4). At r_C under CM, entry after a purchase can take any value in $[\rho, 1]$.

The slot's completeness follows from a finite support enumeration. Each class has three orders, and each gross return falls strictly with the class's own rate.

Research. Only the slot models research. The investor chooses an intensity $z \in [0, \bar{\lambda}]$ at cost $(a/2)z^2$ before it learns the class, and the intensity is the presence probability (Section 5). Zero research is an equilibrium from r_0 to $r_N = 1.747878$, because a unit order against an uninformative price earns $\rho\Delta_T/2 \le k$ there. The equilibrium set with research on $(r_N, r_U^R) = (1.747878, 2.434046)$ is open.

Timing. Both set-ups price one trade before the challenger's decision, and neither models the choice of a trading date. The Laplace flow is the aggregate net demand of the decision interval. The slot is one public order opportunity at a known date. This is a stronger institutional assumption, whose counterpart is the first prespecified opportunity with a recorded zero. In both set-ups the ceiling r_C is a tie event with positive probability, so the challenger's tie rule decides entry at r_C itself.

3.2 Conclusions each set-up supports

The fixed-information control: falls, or does not rise. *Laplace.* Hold full orders fixed and let the challenger decide again. Entry falls strictly from 0.562178 at r_0 to 0.522757 at r_1 (Section 4.2; closed forms). The threshold flow x^* moves continuously with strength, so the crossing probability has a density and the fall is strict.

Slot. With the same control, entry is unchanged at 0.522757 on $(1, r_C]$ and equals ρ above $r_C = 3.592659$ (Proposition A.3 and Table 3; analytical, values computer-assisted). The frozen posteriors are three atoms $m < 1/2 < M$, and the threshold $\tau(r)$ exceeds $5/8$ for $r > 1$. Only the top atom can recruit, so entry is a step function of strength.

The condition is the number of prices. With three prices entry moves in steps, so a strict fall at fixed orders needs the threshold to cross an atom. The weak statement "does not rise" holds in both set-ups.

Reserve deterrence: strict, or flat. *Laplace.* The reserve 1.01 excludes the low class. There, preparation falls from 0.543573 at r_0 to 0.513373 at r_1, with unique continuations at both nodes (Table 5; analytical).

Slot. At the reserve 101/100, entry is 0.522757 at both strengths: full orders are unique at both, and only a purchase recruits at both. Strict deterrence returns at the reserve 31/20, where entry falls from 0.522757 to 0.250000 (Proposition A.10 and Table 5; analytical, values computer-assisted). This reserve exceeds r_0 and excludes the weak incumbent.

The condition is the same step structure. Strict deterrence in the slot needs the stronger incumbent to push the top-posterior profit below the high cost.

The empirical mapping. *Laplace.* Section 7 states three predictions. A stronger prepared bidder raises the probability of further preparation when the price precedes the decision, and weakly lowers it when it does not. A buyer that prepares after a favorable return wins more often. Against a strong prepared bidder, preparation returns to its floor while the price stays informative. The flow is aggregate net demand, so its largest price gap is not a unit spread, and no observable is derived from it.

Slot. The same three predictions hold (Section 8), and the slot adds derived observables. The conditional execution spread is $S = \rho\Delta_T\vartheta + (1-\rho)J\,A(M)$, with $\vartheta = u/(2+u)$ the informed share of nonzero orders (Proposition A.11). It is 0.786299 at r_1, and 90.2% of it is the entry component; the entry component is zero above r_C. Sell impact and volume identify $(\vartheta, \lambda, \beta)$ among admissible slots under full orders, while counts alone leave the informed share in the closed interval $[0.250299, 0.760000]$ under rule IM (Proposition A.12); the upper end needs a zero-profit sale, so rule I0 excludes it. Orders predict the winner among prepared challengers above r_C (Proposition A.13).

The slot's formulas need an observation protocol: the first prespecified public opportunity, with a recorded zero. A sample of first executed trades cannot identify the neutral price effect. The data counterpart of each protocol is open.

4 The proof method

An equilibrium proof has three failure modes. A step can be circular. A step can use a property of the equilibrium before the proof establishes it. A conclusion can hold at one parameter point only.

The method below removes all three. It has five parts, and Section 4.6 applies it to one result in each set-up. Each part is a rule with an instance from one of the two manuscripts.

4.1 The equilibrium as an explicit object

The proof starts from the equilibrium written out as one tuple. The tuple lists the order distribution of each class and the posterior and price after every order or flow. It also lists the challenger's belief at every price on the image, the entry rule at every price and cost, and the bids. The tuple is a proof artifact: each later step may use only what it states.

In the slot manuscript, Definition A.1 fixes the tuple $(\sigma_H, \sigma_L, \mu, P, \bar{\mu}, \zeta)$ with truthful bids. Appendix A.3 displays one tuple for each of the three economies of Proposition 2. Appendix A.4 displays the five forms of the high-class order on the path.

In the Laplace manuscript the components of Proposition 2 sit in four places. The orders are in the statement and the posterior in (A.8). The entry rule and the price are in equation (9), and the price belief in Proposition A.2. The components are complete, and the proof is sound. The

five-part form displays them in one place, with the price function on all of $\mathbb{R}$, its two atoms and their probabilities.

4.2 Verification against the fixed object

The conditions are listed: Bayes consistency, challenger beliefs at each price, entry optimality, competitive prices, investor optimality and truthful bids (and research optimality with research). Each condition is checked by direct computation against the stated tuple. A unilateral deviation holds the candidate's price function and entry rule fixed and changes only the law of the order or flow that reaches them. A comparison across economies solves the equilibrium again in each economy; the two never mix.

Two shortcuts break this part. The first uses price sufficiency before the proof establishes it. The rule: a constructed tuple first shows by direct algebra that its prices differ at different posteriors. Only then does the proof read the challenger's belief from the price. A lemma whose premises include the conditions under proof cannot supply that step.

The slot appendix does this in one paragraph before any condition is checked (Appendix A.2, "Constructed tuples"). The price increment identity uses only the price formula. A stated tuple therefore has distinct prices at distinct posteriors, and condition (E2) then gives the belief at each price. The Laplace manuscript reaches the same point through Proposition A.2, which it proves for every candidate from the entry floor.

The second shortcut is the word "symmetrically". The wrong-sign order of class L in the Laplace proof of Section 4.6 needs its own line. A purchase of size $s > 0$ earns $-s \int f(x-s)A_L(x)\,dx - ks < 0 = \Pi_L(0)$, because the residual A_L is nonnegative.

4.3 Uniqueness through bounds on every profile

A uniqueness claim rests on bounds that hold for every candidate profile, not only for the proposed one. The device that makes this possible is the posterior bound. Under every strategy the posterior after every flow or order lies in $[m(\pi), M(\pi)]$. The score bound gives this in the Laplace set-up, the liquidity floor in the slot. The bound lets the proof bound beliefs before it knows the equilibrium.

Three further bounds follow in order. The zero-cost challenger gives the entry floor $e \geq \rho$ at every price. The floor and the posterior bound give the residual bounds $\rho\kappa\Delta_T \leq A_j \leq \Delta_T$ under every candidate. The residual bound and the trade bound force the orders: zero orders when $\Delta_T < k$, full orders when the lower residual bound exceeds k. Only then does the proof read the posteriors, the entry and the prices of the forced profile.

The proofs of part (iii) show two routes with different hypotheses. The entry-first route bounds every reachable belief below $M(\pi)$ and concludes that entry is ρ in every candidate, before any price inversion. It then forces full orders. It needs the posterior bound at every flow or order, which in the slot is the zero-order bound and $\beta \leq 1/3$.

The orders-first route forces full orders from the bounds at the nonzero orders alone, then reads the three posteriors and excludes high-cost entry at each. It needs no bound at the zero order and holds for every $\beta < 1/2$. The Laplace worked proof uses the first route and the slot appendix the second; both are non-circular.

4.4 Strict margins and the open parameter set

Every hypothesis is a strict inequality, and the proof records its margin at the benchmark in a table. Structural assumptions are listed apart from the strict economic hypotheses: weak dominance between the classes, supports above the reserve, continuity of F, the bound $\beta \leq 1/3$. The open set is stated in named coordinates. A printed interval in one coordinate is a slice through the benchmark with the other coordinates fixed, and the text says so.

The admissible region is then a result. It is a table with each boundary, the benchmark point and its distance to the boundary (slot manuscript, Table 2 and Table A.6). A result on an open set survives a joint perturbation of the primitives, with a radius that the margins bound.

4.5 Certificates for finite steps

A finite step is certified by exact rational arithmetic when its inputs are rational. It is certified by outward interval arithmetic when a transcendental constant enters. Exact decimal inputs are parsed as decimals. A root is certified by a bracket with opposite endpoint signs, and a count of roots by a Sturm sequence. The named script and run log are part of the proof, and the status label is computer-assisted.

The certificate is the exported endpoint, not the sign inside the computation. An interval routine can be exact and still write both endpoints through a nearest-rounded decimal format. Most rows then print equal endpoints, and the true value lies outside the printed pair. The rule: round the lower endpoint down and the upper endpoint up from exact endpoints, or mark a rounded decimal as an approximation.

The slot manuscript applies it. Appendix A.3 names the file of exact rational endpoints as the certificate of each printed value, and the text writes $\approx$ before the prior cut points and the research bounds. A pair of binary floats on either side of a computed value is not a certificate either.

The four thresholds of the Laplace manuscript's Proposition A.4 are closed forms evaluated in floating point, with the two float neighbours as saved bounds. The enclosures that this note prints (Sections 2 and 3.1) certify the digits shown.

4.6 One worked proof in each set-up

The result is part (iii) of Proposition 2 in each manuscript. At the strongest incumbent F_2, the unique trading outcome is full orders, and entry returns to ρ. The Laplace version expands the manuscript's Appendix A.2. That appendix proves part (iii) in three sentences, by reference to Steps 4 and 5 of parts (i) and (ii). The slot version follows Step 5 of Appendix A.3 of the slot manuscript. Each part gives the Laplace proof and then the slot proof.

Hypotheses. *Laplace.* Structural: $G_H \leq G_L$ with both supports above p; F_2 continuous. Strict: (L1) $\Delta_T(F_2) > 0$; (L2) $0 < k < (1 - 1/b)\rho\kappa(\pi)\Delta_T(F_2)$; (L3) $B_{F_2}(M(\pi)) < c_H$; and $b > 1$, $0 < \rho < 1$, $0 < \pi < 1$, $c_H > 0$. In the manuscript, (L1) and (L2) follow from (A2) at F_1 and Proposition 1(b). (L3) is the one new hypothesis of part (iii).

Slot. Structural: the same classes; F_2 continuous; $0 < \beta < 1/2$ (the model's $\beta \leq 1/3$ is not used). Strict: (S1) $0 < k < \rho\kappa(\pi)\Delta_T(F_2)$; (S2) $B_{F_2}(M(\pi)) < c_H$; and $0 < \lambda < 1$, $0 < \rho < 1$, $0 < \pi < 1$, $c_H > 0$. (S1) implies $\Delta_T(F_2) > 0$.

Part 1: the object. *Laplace.* $\sigma_H^* = \delta_{+1}$, $\sigma_L^* = \delta_{-1}$. The posterior on $\mathbb{R}$ is $\mu^*(x) = [1 + \frac{1-\pi}{\pi} \exp\{-(|x+1| - |x-1|)/b\}]^{-1}$. It equals $m(\pi)$ on $x \le -1$, rises strictly on $(-1, 1)$ and equals $M(\pi)$ on $x \ge 1$. The price is $P^*(x) = t_0 + \rho[t_L - t_0 + \Delta_T \mu^*(x)]$, with image $[P_-, P_+]$ and atoms at both ends. The atom P_+ has probability $1/2$ in class H and $\mathrm{e}^{-2/b}/2$ in class L. The belief at a price is the inverse, $\beta^*(P) = (P - t_0 - \rho[t_L - t_0])/(\rho\Delta_T)$. The entry rule is $e^*(P, 0) = 1$ and $e^*(P, c_H) = 0$ at every price, and bids are truthful.

Slot. $\sigma_H = \delta_1$, $\sigma_L = \delta_{-1}$. The posteriors after $(-1, 0, 1)$ are (m, π, M). The prices are $P(x) = t_0 + \rho[t_L - t_0 + \Delta_T \mu(x)]$, three distinct numbers. At the benchmark they are 0.610002, 0.664236 and 0.718471. The belief at each price is the posterior of its one order. The high-cost preparation probability is $\zeta \equiv 0$, so $e \equiv \rho$, and bids are truthful.

Part 2: the conditions. In both set-ups the conditions are:

(C1) Bayes consistency of the posterior and of the belief at each price;

(C2) competitive prices;

(C3) entry optimality at every reached price and cost, with the tie rule;

(C4) investor optimality against the fixed P and e, over pure and mixed orders;

(C5) truthful bids.

The Laplace (C4) ranges over $q \in [-1, 1]$; the slot (C4) over $q \in \{-1, 0, 1\}$.

Part 3: verification. *Laplace.* (C1): under σ^* the flow densities are $f(x-1)$ and $f(x+1)$, and Bayes' rule gives μ^*. P^* is a strictly increasing affine image of μ^*, so $\mu^*(X) = \beta^*(P^*(X))$. (C2): the cost is independent of the class, so $\mathbb{E}[V_T \mid j, x] = t_0 + \rho(t_j - t_0)$; average over the class. (C3): at every reached price the belief lies in $[m, M]$, so $B_{F_2} \le B_{F_2}(M) < c_H$ by (L3). The high-cost challenger stays out, and the zero-cost challenger prepares. (C4): the residuals are $A_H = \rho\Delta_T(1 - \mu^*)$ and $A_L = \rho\Delta_T\mu^*$, both in $[\rho\kappa\Delta_T, \Delta_T]$.

Laplace. For a correctly signed order of size s, $U_j(s) = s\bar{A}_j(s) - ks$ with $\bar{A}_j(s) = \int f(x \mp s) A_j(x)\, dx$. The translation lemma (Laplace manuscript, Online Appendix A.3) gives $|\bar{A}_j'| \le \bar{A}_j/b$. Hence $U_j'(s) \ge (1 - 1/b)\rho\kappa\Delta_T - k > 0$ almost everywhere by (L2), and the unit order strictly dominates every smaller size. A wrong-signed order of size s earns $-s \int f(x \pm s) A_j(x)\, dx - ks < 0$ in each class (Section 4.2). (C5): weak dominance in a second-price auction.

Slot. (C1): the order probabilities are $f_H = (b, b_0, b + \lambda)$ and $f_L = (b + \lambda, b_0, b)$, with $b_0 = (1 - \lambda)(1 - 2\beta)$, and Bayes' rule gives (m, π, M). The three prices differ because $\Delta_T > 0$ and $m < \pi < M$, so each price has one order in its preimage. (C2): the price formula with $e \equiv \rho$. (C3): $B_{F_2}(\mu) \le B_{F_2}(M) < c_H$ at each of the three posteriors by (S2), with no tie. (C4): $K_H = \rho\Delta_T(1 - M) > k$ and $K_L = \rho\Delta_T m > k$ by (S1), so $U_H(1) > 0 = U_H(0) > U_H(-1)$ and $U_L(-1) > 0 = U_L(0) > U_L(1)$. A mixture over other orders averages strictly smaller payoffs. (C5): as in the Laplace set-up.

Part 4: uniqueness through bounds on every profile. *Laplace.* Take any equilibrium with arbitrary mixed orders. Step 1: the triangle inequality gives $\mathrm{e}^{-2/b} \le a_H(x)/a_L(x) \le \mathrm{e}^{2/b}$ for the mixed flow densities. Hence $\mu(x) \in [m, M]$ at every flow and $\beta(P) \in [m, M]$ at every price. Step

Table 3. Margins of Proposition 2(iii) at $r_2 = 18/5$, benchmark, prior $1/2$.

Hypothesis	Margin	Laplace	Slot
Exclusion	$c_H - B_{r_2}(M)$	0.002689	0.002689
Exclusion, posterior units	$\tau(r_2) - M$	0.000334	0.000334
Trade bound	Laplace $(1 - 1/b)\rho\kappa\Delta_T(r_2) - k$; slot $\rho\kappa\Delta_T(r_2) - k$	0.011563	0.043127
Spread	$\Delta_T(r_2)$	0.938889	0.938889
Strengthening	$\Delta_T(r_2) - \Delta_T(r_1)$	0.272222	0.272222
Position above the ceiling	$r_2 - r_C$	0.007341	0.007341
Zero-cost strictness	$g_L(r_2)$	5/48	5/48
Score bound; presence	$b - 1$; λ, $1 - \lambda$	1	0.336122, 0.663878
Floor, prior, cost, fee	ρ, $1 - \rho$, π, $1 - \pi$, c_H, k	1/4, 3/4, 1/2, 1/2, 6, 1/50	the same

Notes. Laplace values are independent outward enclosures of the pinned benchmark; slot values are registry enclosures. The ceiling is $r_C = 3.592659$ in both set-ups (Laplace: (A.11); slot: $\tau(r_C) = M$).

2: by (L3), the high-cost challenger stays out at every reached price, so $e \equiv \rho$ in every equilibrium; this step uses no price inversion. Step 3: with $e \equiv \rho$, $P(x) = t_0 + \rho[t_L - t_0 + \Delta_T\mu(x)]$ for almost every flow, and the residuals under a deviation lie in $[\rho\kappa\Delta_T, \Delta_T]$.

Laplace. Step 4: the derivative bound of Part 3 holds against this profile's own P and e, so both classes use the unit order. Step 5: the orders pin $\mu = \mu^*$, $P = P^*$ up to a flow-null set, $\beta = \beta^*$ on the image and $e \equiv \rho$. The null set is never reached, because every order gives the flow a positive density.

Slot. Take any equilibrium. Step 1: under every strategy $b \le f_j(\pm 1) \le b + \lambda$, so $\mu(1) \le M$ and $\mu(-1) \ge m$. Step 2: the floor $e \ge \rho$ and Step 1 give $K_H \ge \rho\Delta_T(1 - M) \ge \rho\kappa\Delta_T > k$ and $K_L \ge \rho\Delta_T m > k$ by (S1). Every equilibrium therefore has $\sigma_H = \delta_1$ and $\sigma_L = \delta_{-1}$. Step 3: the forced orders give the posteriors (m, π, M). Step 4: by (S2), B_{F_2} lies below c_H at each posterior and at every average of them, so $\zeta \equiv 0$ and $e \equiv \rho$ with no tie. Step 5: the price formula fixes the three prices. Every comparison in Steps 2 and 4 is strict, so the outcome is the same under all four tie-rule pairs.

Part 5: margins and the open set. Table 3 gives the margins at $r_2 = 18/5$ in both set-ups. The auction objects are exact rationals: $g_H(r_2) = 5879/720$, $g_L(r_2) = 5/48$, $\Delta_T(r_2) = 0.938889$ and $\tau(r_2) = 4245/5804$. Because M is the same, the exclusion margin, its posterior form and the position of r_2 above the ceiling coincide. The trade margin differs by the factor $1 - 1/b$.

The open set, as slices through the benchmark, is $r_2 \in (r_C, h)$, $c_H \in (5.997311, 6.217155)$ and $\pi \in (0.467850, 0.500424)$ in both set-ups. The window for k starts at $\Delta_T(r_0) = 0.016667$ in both. It ends 0.002412 above k in the Laplace set-up and 0.024824 above k in the slot. All hypotheses are strict and continuous in the primitives, so the joint region is open and contains the benchmark. Neither tie rule is used at r_2. The zero-cost decision is strict because $g_L(r_2) > 0$, the high-cost decision is strict by the exclusion margin, and both investor comparisons are strict.

Status. Both propositions are analytical, and both margins tables are computer-assisted. The Laplace manuscript's three-sentence proof is sound and non-circular. The five-part form adds the displayed tuple, the margins table, the wrong-sign line for class L and the structural-strict split. It also notes that Step 2 needs no price inversion. The slot manuscript's Appendix A.3 already has the five parts.

4.7 A checklist

A reader can apply these questions to any equilibrium claim.

1. Is the equilibrium displayed as one tuple, with every component on every reachable history?
2. Are the conditions listed, and is each one checked by a computation against that tuple?
3. Does every deviation hold the candidate's price function and entry rule fixed?
4. Does any step use a property of the equilibrium (price sufficiency, a posterior, an entry rate) before the proof establishes it for every candidate?
5. Does the uniqueness argument start from bounds that hold for every profile, and in which order does it force beliefs, residuals, orders, entry and prices?
6. Is every hypothesis a strict inequality with a recorded margin, apart from the structural assumptions?
7. Is the open set stated in named coordinates, with each printed interval identified as a slice?
8. Is every finite step certified by exact or outward arithmetic, with a named script and run log, and are the exported endpoints outward?
9. Which conclusions depend on a tie rule, and is each rule stated?
10. Does every number in the text match its certificate at the printed precision?

5 Where each set-up stands against the standard

The Laplace manuscript. Proposition 2 meets the core of the standard. Each condition is checked against a fixed candidate, and no step assumes its conclusion. Uniqueness rests on the posterior bounds, the entry floor, the residual bounds and the derivative bound, all of which hold for every profile. Eleven compressed steps across the appendices hide no gap. Four points fall short in form:

- no result displays its equilibrium as one tuple;
- the part (iii) margin at the benchmark, $c_H - B_{r_2}(M) = 0.002689$, is not printed (the notes of Table 2 print the four margins of parts (i) and (ii));
- the four thresholds of Proposition A.4 have no certificate (Section 4.5);
- the region is asserted open and proved nonempty, with prior windows printed, but it is not tabulated.

Proposition 3 certifies seven things at each of three strengths:

- an exact root in a stated bracket (at 1.55 the endpoint signs are of order 1.1×10^{-10} and -9.6×10^{-11}: bracket widths, not hypothesis margins);
- the low class's unique best response against the schedule at that root;
- the high class's unique best response against every schedule in the bracket (global lower bounds 7.6×10^{-5}, 0.002753 and 0.005492);
- the exclusion of wrong-signed orders;
- Bayes consistency and competitive prices for the fixed candidate;
- coexistence with no trade (pooling margins 0.007802, 0.005938 and 0.003996; the threshold order $M_v - \tau$ is at least 0.028042 at 1.55);
- the strict order of the three entry enclosures.

It does not certify uniqueness of the equilibrium, a continuous branch between the nodes, or a parameter box around them. Two sentences of Section 4.3 rest on the numerical grid: the sale magnitude rises toward one, and entry rises along the continuation. The manuscript labels both as diagnostics.

The slot manuscript. Appendix A has the five parts for every result. Definition A.1 fixes the tuple and the conditions (E1) to (E6), and Appendix A.5 adds (E7). Each proof displays its tuple. The three economies of Proposition 2 are in A.3, the five forms on the path in A.4, the research tuples in A.5.

Uniqueness starts from the bounds of Lemma A.1, which hold under every profile, and on the path from an enumeration over the finite support. Table A.6 lists every strict hypothesis with its margin at the benchmark, and Table A.3 lists which conclusions depend on a tie rule. Two monotonicity steps on the path rest on exact Sturm counts, and every statement that involves e uses interval arithmetic. These are computer-assisted, the rest analytical.

Three statements depend on the tie rules. The strict rise on $[r_A, r_F]$ needs IM. The threshold pair, with entry 0.341462, 0.321311 and 0.305208 at the three intermediate strengths, and the entry interval at r_C exist only under CM. Under I0 the set is empty on $(r_N, r_S]$. Proposition 2 and the research equilibria at the six benchmark strengths hold under all four pairs.

Margins that are thin in both. Table 4 lists the smallest margins. The return margin at r_2 is the same in both set-ups, and so is the prior window's upper end. In the Laplace manuscript the strong-trade margin, 0.002412, is smaller than the return margin in the same units. The return margin is therefore not the thinnest. In the slot the thin margins are one distance in several coordinates. Margins in different units are not ranked against each other.

6 Decision points

1. **Which set-up carries the main text.** The Laplace set-up keeps the partial sale and three certified nodes with a strict rise. The slot keeps a complete equilibrium set, research, coexistence at every prior and derived observables. Its price is an exogenous liquidity motive and a unit menu. They can also be nested, one as the model and the other as an appendix.

Table 4. The smallest margins at the benchmark.

Margin	Units	Laplace	Slot
Return at r_2: $c_H - B_{r_2}(M)$	cost	0.002689	0.002689
Return at r_2: $\tau(r_2) - M$	posterior	0.000334	0.000334
Prior window, upper end minus $1/2$	prior	0.000424	0.000424
Position above the ceiling, $r_2 - r_C$	strength	0.007341	0.007341
Strong trade bound at r_1	cost	0.002412	0.024824
Weak trade bound, $k - \Delta_T(r_0)$	cost	0.003333	0.003333
Slot band, upper end minus u	u	—	0.004617
Cap strip, upper end minus $\bar{\lambda}$	intensity	—	0.000599
Research cost window around $a = 1/40$	cost parameter	—	0.021469 to 0.036763
Cost-law extension: the margin that binds	cost	0.002412	0.002688902

Notes. The last row reports the margin that binds each cost-law extension. The Laplace strong-trade margin is unchanged by the cost mixture. The slot's return at r_2 survives a high-cost band of half-width below the value shown.

2. **What the strict rise at intermediate strengths must carry.** A sale size (Laplace, three nodes); a mixture probability (slot, $[r_A, r_F]$, rule IM); or a research intensity (slot, $[1.501023, 1.677633)$, a quadratic cost and a cap).
3. **The fixed-information control and reserve deterrence.** The Laplace set-up supports "falls strictly" and strict deterrence at the exclusion reserve. The slot supports "does not rise" and flat entry at that reserve, with strict deterrence at 31/20.
4. **The thin boundary at r_2.** Both set-ups sit 0.000334 below the ceiling in posterior units. A larger r_2, or a larger c_H that stays below $B_{r_1}(M)$, widens this margin in both.
5. **Tie rules in the statements.** The slot results name IM and C1 and list where I0 and CM change a conclusion. The Laplace manuscript uses the challenger's rule at r_C and at two reserve events, and needs no investor rule.
6. **The five-part form in the Laplace manuscript.** Four additions bring it to the standard: an explicit tuple for Proposition 2, the part (iii) margin, outward certificates for the four thresholds and a tabulated region.
7. **Open items that a reader will ask about.** Laplace: the first informative equilibrium, the set between r_N and r_U, a certified branch. Slot: the set with research on (r_N, r_U^R), the selection between the two outcomes on $[r_A, r_N]$ and the data counterpart of each observation protocol. Both: the seller's reserve with the full continuation and the trading game under the bargaining rule.

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